

1



Mark for Review



The value of  $y$  is 8 more than twice the value of  $x$ . Which equation represents  $y$  in terms of  $x$ ?

(A)  $y = 10x + 2$

(B)  $y = 2x + 8$

(C)  $y = 10x + 8$

(D)  $y = 8x + 2$



2



Mark for Review



To purchase a used car at a total price of 11,000 dollars, a onetime down payment is required, and then fixed monthly payments are made for the remaining amount owed for the car. The equation  $11,000 = 2,200 + 200t$  represents this situation, where  $t$  is the number of fixed monthly payments that are made. Which of the following is the best interpretation of 200 in this context?

- (A) The amount, in dollars, of the down payment
- (B) The amount, in dollars, of each fixed monthly payment
- (C) The total amount, in dollars, paid for the car after  $t$  fixed monthly payments
- (D) The total number of fixed monthly payments

3



Mark for Review



Line  $k$  is defined by  $y = 3x - 2$ . Line  $j$  is parallel to line  $k$  in the  $xy$ -plane and passes through the point  $(0, 1)$ . Which equation defines line  $j$ ?

(A)  $y = x - 2$

(B)  $y = 3x + 1$

(C)  $y = -x - 2$

(D)  $y = -3x + 1$



4



Mark for Review



How many solutions does the equation  $5x + 3 = 5x$  have?

☐ (A) Zero

☐ (B) Exactly one

☐ (C) Exactly two

☐ (D) Infinitely many

5



Mark for Review



A model  $\bar{p}$  predicts that the population of a certain town was 5,000 in 2008. The model also predicts that each year for the next 5 years, the population  $p$  increased by 4% of the previous year's population. Which equation best represents this model, where  $x$  is the number of years after 2008, for  $x \leq 5$ ?

(A)  $p = 0.96(5,000)^x$

(B)  $p = 1.04(5,000)^x$

(C)  $p = 5,000(0.96)^x$

(D)  $p = 5,000(1.04)^x$

6



Mark for Review



A park rents beach umbrellas to visitors. The park earns revenue of \$15 for each beach umbrella they rent for the day. On Wednesday, the park earned \$435 in profit from renting beach umbrellas after paying daily expenses of \$150. How many beach umbrellas did the park rent on Wednesday? (profit = total revenue – total expenses)

(A) 19

(B) 29

(C) 38

(D) 39

7



Mark for Review



$$r = \sqrt{9w - 5}$$

The given equation relates the real numbers  $r$  and  $w$ , where  $w > \frac{5}{9}$ . Which equation correctly expresses  $w$  in terms of  $r$ ?

(A)  $w = \sqrt{-5r + 9}$

(B)  $w = \sqrt{9r - 5}$

(C)  $w = \frac{r^2}{9} + \frac{5}{9}$

(D)  $w = \frac{r^2}{9} + 5$





A hemisphere is half of a sphere. If a hemisphere has a radius of 83 inches, which of the following is closest to the volume, in cubic inches, of this hemisphere?

(A) 14,400

(B) 57,700

(C) 898,200

(D) 1,197,500



Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as  $3\frac{1}{2}$  ), enter it as an improper fraction ( $7/2$ ) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

| Answer | Acceptable ways to enter answer | Unacceptable: will NOT receive credit |
|--------|---------------------------------|---------------------------------------|
| 3.5    | 3.5<br>3.50<br>7/2              | 3 1/2<br>3 1/2                        |
|        |                                 |                                       |

9

Mark for Review

For a certain type of rope, the equation  $y = 900ax^2$ , where  $a$  is a constant, gives the estimated breaking strength  $y$ , in pounds, of a rope with a circumference of  $x$  inches. Based on this equation, if a rope of this type has a circumference of **0.75** inches, it has an estimated breaking strength of **708.75** pounds. What is the estimated breaking strength, in pounds, of a rope of this type that has a circumference of **4.50** inches?

Answer Preview:

10



Mark for Review



The function  $h$  is defined by  $h(x) = (x + p)(x - 4)(2x - 6)$ , where  $p$  is a constant. In the  $xy$ -plane, the graph of  $y = h(x)$  passes through the point  $(-6, 0)$ . What is the value of  $h(0)$ ?

(A) 144

(B) 1

(C) -6

(D) -72



## Student-produced response directions

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### Examples

| Answer | Acceptable ways to enter answer | Unacceptable: will NOT receive credit |
|--------|---------------------------------|---------------------------------------|
| 3.5    | 3.5<br>3.50<br>7/2              | 3 1/2<br>3 1/2                        |
|        |                                 |                                       |

11



Mark for Review

If  $x \geq -8$  represents all solutions to the inequality  $ax - 18 \leq 14$ , where  $a$  is a constant, what is the greatest possible value of  $a$ ?

### Answer Preview:

12



Mark for Review



$$5(3x) + 4(8y) = 16$$

$$-5(3x) + 4(8y) = 16$$

The solution to the given system of equations is  $(x, y)$ . What is the value of  $3x + 8y$ ?

(A) 0.5

(B) 0.8

(C) 4

(D) 16



## Student-produced response directions

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- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

### Examples

| Answer | Acceptable ways to enter answer | Unacceptable: will NOT receive credit |
|--------|---------------------------------|---------------------------------------|
| 3.5    | 3.5<br>3.50<br>7/2              | 3 1/2<br>31/2                         |
|        |                                 |                                       |

13



Mark for Review

For the positive quantities  $h$ ,  $j$ , and  $k$ , 35% of  $h$  is equivalent to 49% of  $j$ , and  $j$  is equivalent to 45% of  $k$ . What percentage of  $k$  is  $h$ ? (Disregard the % sign when entering your answer. For example, if your answer is 39%, enter 39)

### Answer Preview:

## Student-produced response directions

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## Examples

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|--------|---------------------------------|---------------------------------------|
| 3.5    | 3.5<br>3.50<br>7/2              | 3 1/2<br>3 1/2                        |
|        |                                 |                                       |

14



Mark for Review

$$4x^2 + 19 = 32x + r$$

In the given equation,  $r$  is a constant. The equation has exactly one real solution. What is the value of  $r$ ?

## Answer Preview:



15



Mark for Review



For the function  $f$ , for every increase of 2 in the value of  $x$ , the value of  $f(x)$  increases by a factor of  $c$ , where  $c$  is a constant. Which of the following equivalent forms of function  $f$  displays the value of  $c$  as the base or the coefficient?

(A)  $f(x) = 48(2)^{6x}$

(B)  $f(x) = 48(8)^{2x}$

(C)  $f(x) = 48(64)^x$

(D)  $f(x) = 48(4,096)^{\frac{x}{2}}$

## Student-produced response directions

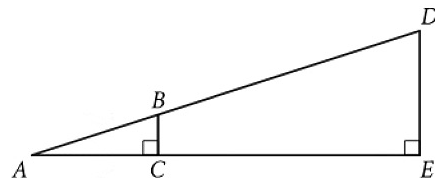
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## Examples

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|--------|---------------------------------|---------------------------------------|
| 3.5    | 3.5<br>3.50<br>7/2              | 3 1/2                                 |

16

Mark for Review



Note: Figure not drawn to scale.

In the figure shown,  $AB = \sqrt{61}$  units,  $AC = 6$  units, and  $CE = 18$  units. What is the area, in square units, of triangle  $ADE$ ?

Answer Preview:

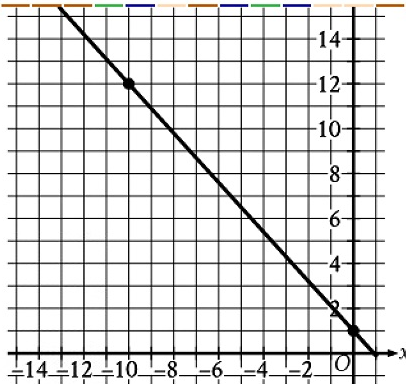
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- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

## Examples

| Answer | Acceptable ways to enter answer | Unacceptable: will NOT receive credit |
|--------|---------------------------------|---------------------------------------|
| 3.5    | 3.5<br>3.50<br>7/2              | $3\frac{1}{2}$<br>$3\frac{1}{2}$      |

17.



The graph of line  $g$  is shown in the  $xy$ -plane. Line  $k$  is defined by  $165x + py = w$ , where  $p$  and  $w$  are constants. If line  $k$  is graphed in this  $xy$ -plane, resulting in the graph of a system of two linear equations, the system of two linear equations will have infinitely many solutions. What is the value of  $p + w$ ?

Answer Preview:



18



Mark for Review



Each of the following frequency tables represents a data set. Which of these frequency tables represents the data set with the smallest standard deviation?

A.

| Value | Frequency |
|-------|-----------|
| 5     | 0         |
| 6     | 10        |
| 7     | 55        |
| 8     | 10        |
| 9     | 0         |

C.

| Value | Frequency |
|-------|-----------|
| 5     | 15        |
| 6     | 15        |
| 7     | 15        |
| 8     | 15        |
| 9     | 15        |

B.

| Value | Frequency |
|-------|-----------|
| 5     | 11        |
| 6     | 17        |
| 7     | 19        |
| 8     | 17        |
| 9     | 11        |

D.

| Value | Frequency |
|-------|-----------|
| 5     | 18        |
| 6     | 14        |
| 7     | 11        |
| 8     | 14        |
| 9     | 18        |



19



Mark for Review



$$x^2 - 4x + y^2 - 8y - 80 = 0$$

In the  $xy$ -plane, the graph of the given equation is a circle. If this circle is inscribed in a square, what is the perimeter of the square?

(A) 20

(B) 40

(C) 80

(D) 320



20



Mark for Review



$$7x + 5y = 2$$

The given equation is one equation in a system of two linear equations. If the system of equations has at least one solution, which of the following equations could be the other equation in the system?

I.  $10.5x + 7.5y = 3$

II.  $10.5x - 7.5y = 3$

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II

21



Mark for Review



The function  $f$  is defined by  $f(x) = ax^2 + bx + c$ , where  $a$ ,  $b$ , and  $c$  are constants. The graph of  $y = f(x)$  in the  $xy$ -plane passes through the points  $(8, 0)$  and  $(-2, 0)$ . If  $a$  is an integer greater than 1, which of the following could be the value of  $a + b$ ?

(A)  $-10$

(B)  $-5$

(C)  $6$

(D)  $7$



22



Mark for Review



An object's speed is increasing at a rate of 2.7 meters per second squared. What is this rate, in miles per minute squared, rounded to the nearest tenth? (Use 1 mile = 1,609 meters.)

(A) 0.1

(B) 6

(C) 72.4

(D) 595.9